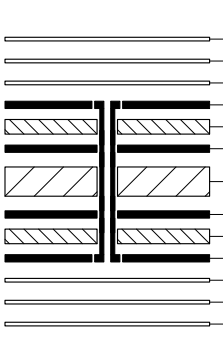
Tessera Badge Fabrication Document

Layer Stack Legend



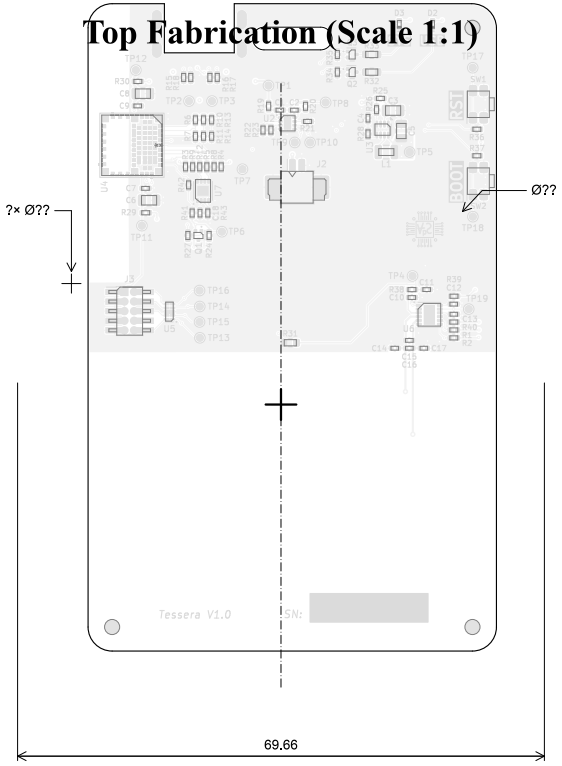
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.1164mm	FR4_2116	Dielectric	
Copper	**L2 (Sig, PWR)**	**0.0152mm (0.43oz)**		**Signal**	**GBR**
Core		0.665mm	FR4	Dielectric	
Copper	**L3 (GND)**	**0.0152mm (0.43oz)**		**Plane**	**GBR**
Prepreg		0.1164mm	FR4_2116	Dielectric	
Copper	**L4 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.0382mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL SNPB

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY APS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	54.000 × 85.610 mm
BOARD THICKNESS	1.038 mm
TRACE WIDTH	0.120 mm
TRACE TO TRACE	0.100 mm
MIN. HOLE (PTH)	0.200 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.050 mm
COPPER TO HOLE	0.150 mm
COPPER TO EDGE	0.200 mm
HOLE TO HOLE	0.254 mm

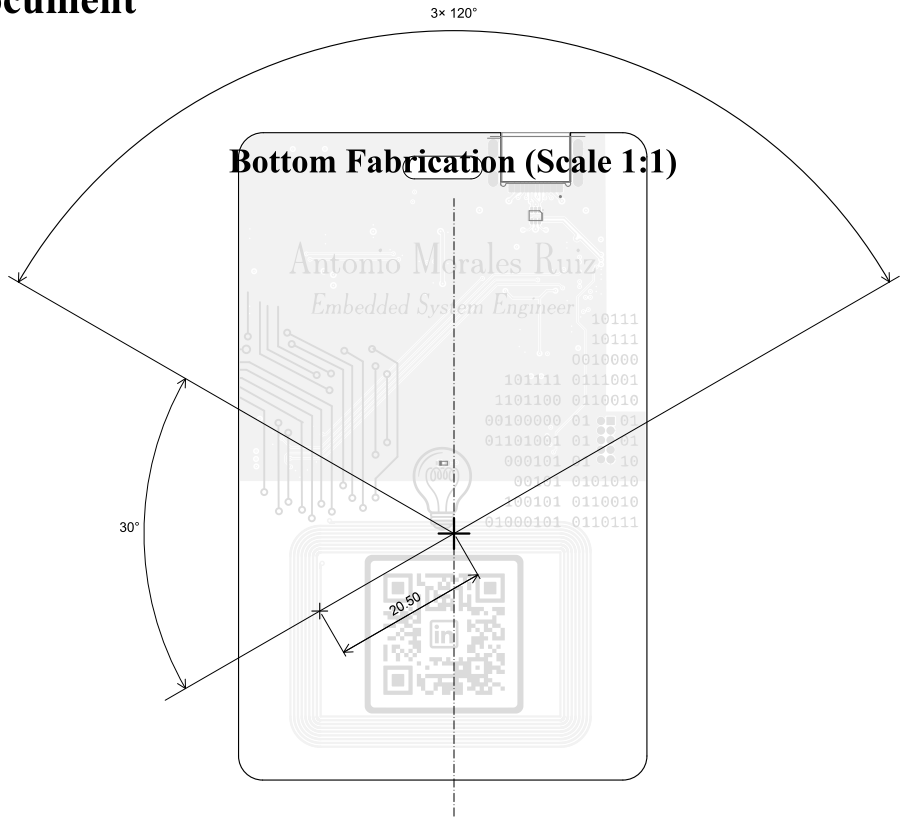
11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **1** of **8**

Tessera Badge Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

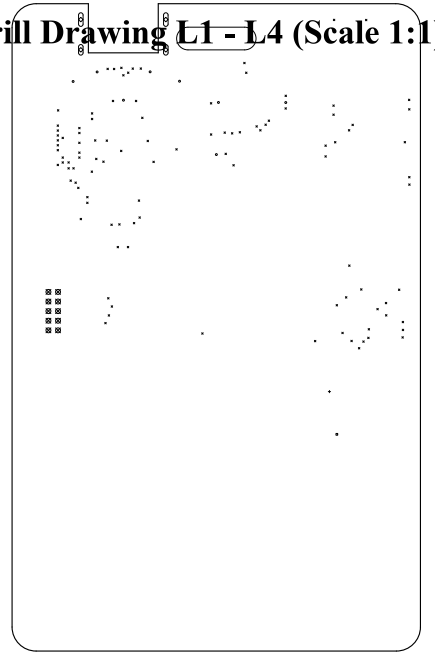
	Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
		Board Name: **Tessera Badge**		Project Name: **Tessera**	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: **A4**	Sheet: **2** of **8**

Tessera Badge Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	105	0.20mm (7.87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
○	8	0.25mm (9.84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	1	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	1	0.33mm (12.99mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	4	0.60mm (23.62mils)	PTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⊠	10	0.65mm (25.59mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
	Total 129					

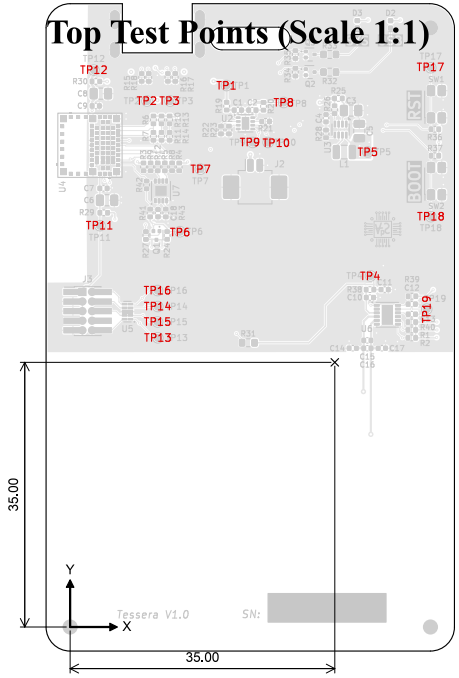
Drill Drawing L1 - L4 (Scale 1:1)



	Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
		Board Name: Tessera Badge		Project Name: Tessera	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 8

Tessera Badge Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP1	VBUS	20.68	71.60
TP2	D-	10.16	69.55
TP3	D+	13.21	69.55
TP4	VCC_EH	39.73	46.42
TP5	VCC	39.34	62.81
TP6	VCC_SEC	14.52	52.28
TP7	GND	17.23	60.54
TP8	VSYS	28.23	69.37
TP9	CHG_STAT	24.15	64.08
TP10	CHG_PG	26.18	64.08
TP11	VCC_nRF	3.94	53.08
TP12	VBUS_nRF	3.14	73.65
TP13	RST_PB	11.60	38.24
TP14	SWDCLK	11.59	42.38
TP15	SWDO	11.56	40.34
TP16	SWDIO	11.56	44.49
TP17	RST_PB	47.68	73.96
TP18	USR_PB	47.70	54.27
TP19	NFC_INT	47.16	42.04

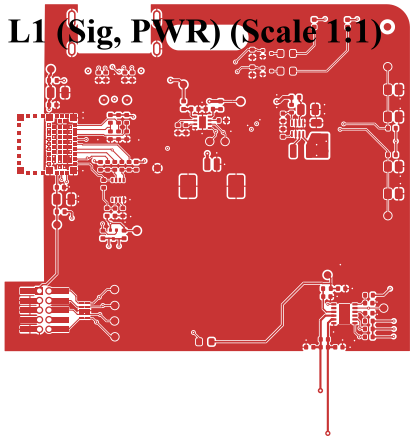
Ref.	Net	X [mm]	Y [mm]

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
		Board Name: **Tessera Badge**		Project Name: **Tessera**	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: **A4**	Sheet: **4** of **8**

Tessera Badge Fabrication Document

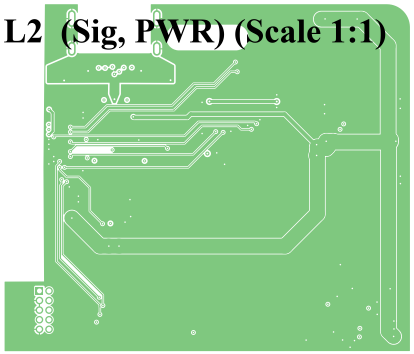
L1 (Sig, PWR) (Scale 1:1)



	Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
		Board Name: **Tessera Badge**		Project Name: **Tessera**	
	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: **A4**	Sheet: **5** of **8**

Tessera Badge Fabrication Document

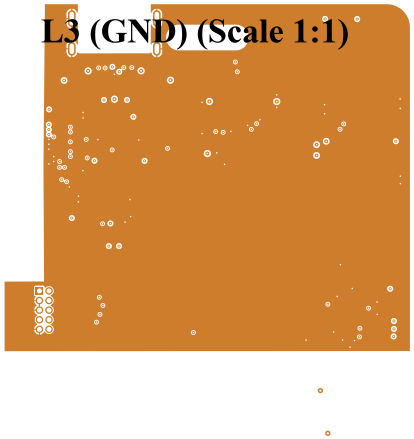
L2 (Sig, PWR) (Scale 1:1)



	Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
		Board Name: **Tessera Badge**		Project Name: **Tessera**	
	Sheet Title: L2 (Sig, PWR) (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: **A4**	Sheet: **6** of **8**

Tessera Badge Fabrication Document

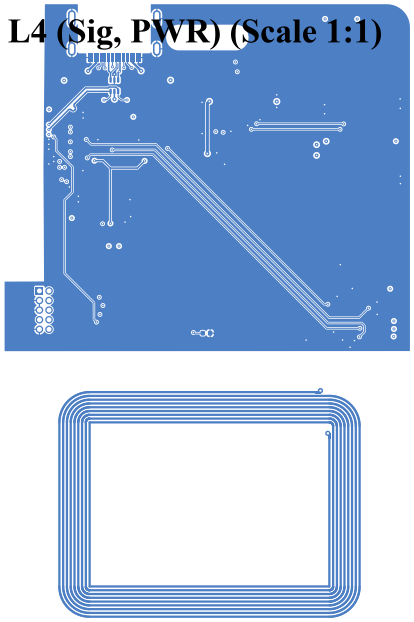
L3 (GND) (Scale 1:1)



	Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
		Board Name: **Tessera Badge**		Project Name: **Tessera**	
	Sheet Title: L3 (GND) (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: **A4**	Sheet: **7** of **8**

Tessera Badge Fabrication Document

L4 (Sig, PWR) (Scale 1:1)



Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 945b98e
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **8** of **8**